

GENERAL THETA FOUNDATIONS I: INTRINSIC CAUSAL DEFICIENCY AND FINITE-STATE DECISIONS

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ABSTRACT. We study the least marked feedback simulation error at a prescribed persistent-state and randomization budget. The finite theory distinguishes private, hidden-selector, and visible-selector deficiencies. Hidden convexification has a testing dual with a priced finite realization; it need not preserve the private zero set. For every fixed private or finite-selector budget we give instead a complete system of local testing certificates and executable upper witnesses. Their error bound depends on the depth and row alphabets, not the number of histories, and they retain the original state budget. The resulting spectra compose and transport endogenously controlled common-encoder decisions. A regenerative theorem treats nonsummable average risks with singular physical transitions. For active hard-sphere dynamics we combine graph-core approximation with finite Gram residual bounds to control the actual marked experiment before consumer optimization. This gives explicit a posteriori risk bounds at unchanged consumer widths, including a finite-tail average-risk bound.

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1. INTRODUCTION

A causal simulation has two outputs: the report delivered to its consumer and the state that must survive until the next report. Classical comparison records the first and may leave the second unpriced. We keep both. The quantity of interest is the least error attainable by one task- and parameter-independent simulator at a prescribed persistent-state budget. The error compares the entire feedback transcript together with its actual latent mark. In particular, equality of report marginals does not replace equality of a marked experiment.

This distinction leads to three optimization problems. A private simulator retains only its declared state and uses fresh randomness. A hidden-selector simulator first selects a policy but does not reveal that choice to the consumer. A visible-selector simulator reveals it and includes it in the compared joint interface. Hidden convexification has a finite testing dual, but its realization may require storing a selector index. At a fixed private width this is not a harmless convention: a delayed three-symbol channel has zero hidden-selector deficiency and strictly positive private deficiency at width two. If the selector is visible, its optimized zero set is again the private zero set.

1.1. The main statement. For finite marked causal experiments E, F , prescribed widths $K = (K_0, \dots, K_T)$ and a fixed finite seed signature, let $\delta_K(E, F)$ be the infimum of the marked feedback error over all admissible simulators. The definitions and exact signature-dependent variational formulas are given in Section 2. The central result can be stated as follows.

At each fixed private or finite-selector budget, intrinsic causal deficiency has a complete family of finite local testing certificates and original-budget upper witnesses. These bounds converge with an explicit causal modulus. They compose across stateless marked presentation changes without increasing the budget, and active microscopic approximation supplies such changes with finite graph-residual bounds.

More precisely, Theorem 2.7 constructs rational lower and upper bounds $L_M \leq \delta_K \leq U_M$ from clipped-simplex policy cells. In the private case, with at most d_t joint output–state labels in a row at time t , their gap is at most

$$1 - \prod_{t=0}^T (1 - \min\{1, d_t/(2M)\}),$$

with a zero contribution from one-point rows. The modulus depends on depth and row alphabets, not on the number of histories. Enumeration of the cells can nevertheless be large; no polynomial-time synthesis conclusion is drawn. The lower bound assigns a marked feedback test to each cell. The upper bound is a single admissible row table at the original budget. This is a nonlinear testing converse, not an unpriced application of convex separation. For dyadic M the bounds are monotone and converge. Strict feasible and infeasible thresholds have finite witnesses; exact boundary decisions use classical real algebraic elimination.

The optimized spectra have the composition law

$$\delta_{KL}(E, G) \leq \delta_K(E, F) + \delta_L(F, G),$$

with selector costs multiplied when present. Theorem 2.9 then transfers an entire intrinsic interval across two-sided stateless presentation changes. Its lower statement excludes all original-budget simulators, even when the original carriers are nonfinite. Its upper statement constructs a simulator

by composing the rational row table with the bridge channels. Thus the scope of an assertion is visible in both its error and its persistent resources.

The other parts of the article give operational and analytic realizations of this comparison. Theorem 3.3 identifies the exact common-encoder face for endogenously controlled finite tasks by reachable compatible covers, and relates the lossy problem to Bellman regrets. Theorem 4.1 treats nonsummable average criteria for stationary finite-register controllers with actual simultaneous physical and register resets; deterministic or singular physical transitions are permitted between resets. Theorem 5.2 supplies active hard-sphere interventions and convergent graph-core approximations. Theorem 5.5 strengthens their finite-horizon comparison by an a posteriori bound involving only graph and intervention Gram data. Its root-sum-square word estimate retains the actual mark and precedes optimization over every finite consumer width. A finite age cutoff adds an explicit geometric remainder for average risks.

1.2. Comparison with classical results. Blackwell–Le Cam comparison, convex minimax, and finite-dimensional convex representation are classical ingredients [1, 2]. Filtered comparison is not new as a subject. Weisshaupt’s filtered stochastic-operator compactness lemma and randomization theorem [10, Lemma 1 and Theorem 5] give an explicit nearby convex comparison result with a finitely additive target space. They do not impose the persistent-state budget considered here. We distinguish that retrieved statement from Norberg’s closely related filtered comparison paper [3]; the full original-source comparison with the latter is not claimed complete.

The common-information dynamic program of Nayyar–Mahajan–Teneketzis [4, Theorem 3 and Corollary 5] provides a classical coordinator formulation, including a finite-memory specialization. The compatible-cover device also has classical incomplete-machine antecedents [12]. Our exact task theorem adds the common encoder, policy-dependent reachable support and Bellman regret conditions; the cover terminology by itself is not a novelty claim. Real quantifier elimination is used only for an exact decision corollary [11]. Similarly, graph-core semigroup approximation and the well-defined microscopic hard-sphere flow are credited to their classical sources [9, 6]. The residual proof below uses a finite-dimensional variation-of-constants formula and does not propose a new abstract semigroup approximation theorem.

The mathematical distinction retained after those ingredients are removed is the simultaneous original-budget testing and realization statement: local feedback tests certify the private feasible set that a convex filtered comparison can enlarge, and the same optimized quantity accepts active microscopic error bounds without a hidden consumer-memory charge. The finite delayed-channel separation makes that distinction explicit. It is not a claim that all ingredients, or their entire conjunction, have an exhaustively certified priority. The source-level literature audit records which original theorems were inspected and which remain to be compared.

1.3. Organization and scope. Section 2 develops the signature-indexed deficiency, its local certificates and its composition-stability theorem. The next sections treat endogenous tasks, regenerative average criteria, and active microscopic realization. All five sections include the proofs they use. The complete development retains the preceding mathematical bodies, including dominated finite-prefix and summable infinite-time duality, marked continuation quotients, exact synthesis, operator memory, Gaussian resource laws, and the historical model interfaces.

The physical results fix particle number, action alphabet, observation noise, and the initial-mark channel. They do not imply the historical nonlinear kinetic action-sublevel corrector or a Boltzmann–Grad limit. Likewise an intrinsic lower certificate is a mathematical inequality over a specified finite model, not an assertion that floating-point Gram entries certify a physical system. The eleven-component program is a mathematical dependency graph with adapters and independent lines; repository chronology and its separately frozen audit are not hypotheses of any theorem in this article.

2. INTRINSIC CAUSAL DEFICIENCY

2.1. Experiments and executable signatures. Fix a finite parameter set Θ , a horizon T , and finite control alphabets $A_0, \dots, A_{T-1}$. An experiment has a report at checkpoints $0, \dots, T$ and an unobserved terminal mark W in a common finite set. It is specified by finite causal instruments: for each parameter and control history they give the joint law of the report path and the actual mark. Equivalently, a finite latent-state realization may be expanded into its history tree. The parameter and mark are unavailable to the encoder. The mark may depend on the physical controlled path. We compare experiments with the same control interface and mark space.

Write Y_t for the source report and Z_t for the target report. A private simulator of width K has stochastic rows

$$\begin{aligned} q_0(z, m \mid y), \quad m \in [K_0], \\ q_t(z, m \mid m', a, y), \quad m' \in [K_{t-1}], \quad m \in [K_t], \quad 1 \leq t \leq T. \end{aligned} \quad (2.1)$$

A row consumes the current source report, emits the target report, and retains the new state. The previous target reports are not available to that row unless encoded in m' . The clock and current control are part of the declared interface. Fresh private randomness is permitted, but no other persistent random variable is free. The product of all row simplexes is denoted $\mathcal{K}_K$.

A deterministic feedback tree g chooses a_t from $z_{0:t}$ and its own previous controls. It is the same controller on both sides of a comparison. Let Ω_g be the finite marked transcript space, including reports, controls, and W . Put

$$P_{\theta,g}^q = \text{Law}_\theta(E \xrightarrow{q} g), \quad Q_{\theta,g} = \text{Law}_\theta(F \rightarrow g).$$

Here W on the left is the source's actual mark. The total variation convention is $\|P - Q\|_{\text{TV}} = \sup_B |P(B) - Q(B)|$. Private deficiency is

$$\delta_K^p(E, F) = \min_{q \in \mathcal{K}_K} \max_{\theta, g} \|P_{\theta,g}^q - Q_{\theta,g}\|_{\text{TV}}. \quad (2.2)$$

The minimum will be justified below. Randomized feedback need not be added to the maximum: conditioning on its independent random table expresses its law as a mixture of deterministic feedback laws. The same argument permits arbitrary independent side randomness in the tester.

A *hidden selector* $U \in [r]$, independent of the source, chooses one simulator q^u . It is not visible to the feedback controller or the final test. A *visible selector* is revealed before feedback begins. The two costs are

$$\delta_{K,r}^h(E, F) = \min_{\lambda, q^1, \dots, q^r} \max_{\theta, g} \left\| \sum_u \lambda_u P_{\theta,g}^{q^u} - Q_{\theta,g} \right\|_{\text{TV}}, \quad (2.3)$$

$$\delta_{K,r}^v(E, F) = \min_{\lambda, q^1, \dots, q^r} \max_{\theta, (g_u)} \left\| \sum_u \lambda_u \delta_u \otimes P_{\theta, g_u}^{q^u} - \sum_u \lambda_u \delta_u \otimes Q_{\theta, g_u} \right\|_{\text{TV}}. \quad (2.4)$$

In both expressions $\lambda \in \Delta_r$ and $q^u \in \mathcal{K}_K$. The second formula compares the joint seed-marked interface. Its target seed has the same distribution and is independent of the target experiment. A hidden selector can be stored privately by retaining (u, m) , at width rK_t . A visible selector is separate side information and is charged by its own alphabet size. Neither interpretation identifies an arbitrary real-valued design with a free finite register.

Definition 2.1 (Deficiency spectrum). The private resource spectrum of E relative to F is the upward-closed set of pairs (K, ε) satisfying $\delta_K^p(E, F) \leq \varepsilon$. The hidden spectrum consists of triples (K, r, ε) satisfying $\delta_{K,r}^h(E, F) \leq \varepsilon$, and the visible spectrum uses $\delta_{K,r}^v$. Width profiles are ordered coordinatewise; in the seeded spectra the seed bound and error are ordered as well. Unused states or selector values may be adjoined. The hidden convexification is $\delta_K^{\text{mix}} = \inf_{r \geq 1} \delta_{K,r}^h$.

The spectrum, not a presumed unique coordinatewise minimum profile, is the intrinsic object. For example one may minimize $\max_t K_t$, with a specified seed signature and tolerance, over its feasible profiles. Different checkpoints can have incomparable optimal tradeoffs.

2.2. The finite comparison theorem. Concatenate all probabilities $P_{\theta,g}^q(\omega)$ into $b(q) \in \mathbb{R}^N$, retaining their row indices; put $b_F = (Q_{\theta,g}(\omega))$. Let $\mathcal{D}_K$ be the finite set of deterministic choices of all rows in (2.1). Empty or unreachable rows remain harmless coordinates. Set

$$a_K = \dim \text{aff}\{b(q) : q \in \mathcal{K}_K\}, \quad \mathcal{I} = \{(\theta, g, B) : B \subseteq \Omega_g\}.$$

For $i = (\theta, g, B)$ and $d \in \mathcal{D}_K$, define

$$A_{i,d} = Q_{\theta,g}(B) - P_{\theta,g}^d(B). \quad (2.5)$$

Including both an event and its complement includes both signs.

Theorem 2.2 (Intrinsic deficiency, priced converse, and visibility). *For the finite marked causal experiments above the following assertions hold.*

(i) *Private deficiency and both fixed- r deficiencies are attained. The private problem has the exact polynomial formulation*

$$\min\{\varepsilon : q \in \mathcal{K}_K, 0 \leq \varepsilon \leq 1, Q_{\theta,g}(B) - P_{\theta,g}^q(B) \leq \varepsilon \text{ for all } (\theta, g, B) \in \mathcal{I}\}. \quad (2.6)$$

In particular its zero set is exactly the nonnegative causal factorization of the target marked behavior by the rows (2.1). At $K_t = 1$ for every t , this is precisely the same factorization with rows $q_t(z \mid a, y)$ and $q_0(z \mid y)$.

(ii) *Hidden convexification has the attained dual*

$$\delta_K^{\text{mix}}(E, F) = \min_{\lambda \in \Delta(\mathcal{D}_K)} \max_{i \in \mathcal{I}} \sum_d \lambda_d A_{i,d} = \max_{w \in \Delta(\mathcal{I})} \min_{d \in \mathcal{D}_K} \sum_i w_i A_{i,d}. \quad (2.7)$$

An optimal simulation uses at most $a_K + 1$ selector values. Thus every weighted causal test has optimal discrepancy at most ε if and only if a hidden mixture of K -state simulators has error at most ε ; it has a private realization of width at most $(a_K + 1)K$. More specifically,

$$\delta_{rK}^p \leq \delta_{K,r}^h, \quad \delta_K^{\text{mix}} = \delta_{K,a_K+1}^h \leq \delta_K^p. \quad (2.8)$$

(iii) *Put $e_\theta(q) = \max_g \|P_{\theta,g}^q - Q_{\theta,g}\|_{\text{TV}}$. Then*

$$\delta_{K,r}^v = \min_{\lambda, q^1, \dots, q^r} \max_{\theta} \sum_u \lambda_u e_\theta(q^u). \quad (2.9)$$

Writing $a_K^v = \dim \text{aff}\{(e_\theta(q))_\theta : q \in \mathcal{K}_K\}$, the infimum over all finite visible selectors is attained using $a_K^v + 1 \leq |\Theta| + 1$ values and equals

$$\max_{v \in \Delta(\Theta)} \min_{q \in \mathcal{K}_K} \sum_{\theta} v_\theta e_\theta(q). \quad (2.10)$$

For every $r \geq 1$, and also after optimizing over all r ,

$$\delta_{K,r}^v(E, F) = 0 \iff \delta_K^p(E, F) = 0. \quad (2.11)$$

No corresponding equivalence with $\delta_K^{\text{mix}} = 0$ holds in general.

(iv) *For compatible experiments E, F, G and profiles K, L ,*

$$\begin{aligned} \delta_{KL}^p(E, G) &\leq \delta_K^p(E, F) + \delta_L^p(F, G), \\ \delta_{KL,rs}^s(E, G) &\leq \delta_{K,r}^s(E, F) + \delta_{L,s}^s(F, G), \quad s \in \{h, v\}. \end{aligned} \quad (2.12)$$

Here KL is the pointwise product, and the second line uses independent selectors. If an F -consumer has widths S_t , composing it with a private simulator gives widths $K_t S_t$ and changes every $[0, 1]$ marked risk by at most $\delta_K^p(E, F)$. A hidden selector is stored at the additional price r , whereas

visible selector alphabets multiply. The same bounds hold for optimized upper risk sets and their lower support functions with the indicated costs.

Proof. Expand a probability of a complete source path, simulator state path, and transcript. It is a product of fixed experiment probabilities and one chosen row entry per time. Summing latent and state paths gives a polynomial in the finitely many row entries. No row belonging to a fixed time can occur twice along one path. Consequently the polynomial is multiaffine in the groups of stochastic rows. Continuity and compactness of the product of closed finite simplexes prove attainment in (2.2); the same proof includes λ and the r policies in (2.3). For the visible problem there are only finitely many collections (g_u) and transcript events, so its objective is again continuous on a compact set. The finite-event formula for total variation proves (2.6). Setting $\varepsilon = 0$ forces equality of every transcript probability, in both directions of each event inequality. This proves the factorization and stateless claims, including their necessity.

For each stochastic row of q , sample its deterministic output once, independently of all other rows, before the experiment. This constructs a probability Λ_q on $\mathcal{D}_K$. On any fixed path the row variables encountered have different time indices, so the expectation of their product is the product of the row probabilities. Summing paths shows, simultaneously for every θ, g , that

$$b(q) = \sum_{d \in \mathcal{D}_K} \Lambda_q(d) b(d). \quad (2.13)$$

This is a behavior identity, not a same-width deterministic purification of minimax error. It implies $\text{conv } b(\mathcal{K}_K) = \text{conv} \{b(d) : d \in \mathcal{D}_K\}$. The latter is a compact polytope. Minimizing the continuous convex maximum of event discrepancies over this polytope gives the first expression in (2.7). Finite-dimensional minimax gives the second, with both optima attained. A point of this polytope is a convex combination of at most $a_K + 1$ of its generating points by Carathéodory's theorem. For completeness, if a representation uses more points, their affine dependence gives coefficients c_d with $\sum c_d = 0$ and $\sum c_d b(d) = 0$. Subtract the largest permissible multiple of c from the positive mixing weights; a weight becomes zero without changing the represented point. Repeating gives the stated bound. Store its selector together with the machine state to obtain (2.8). The dual inequality for every w is equivalent to the maximum over w being at most ε , which proves the priced testing converse.

For disjoint seed fibers, total variation is the sum of the fiber total variations weighted by λ_u . For a fixed θ , the tester can choose a separate maximizing tree on each revealed fiber. Therefore its maximum is $\sum_u \lambda_u e_\theta(q^u)$, proving (2.9). The map $q \mapsto e(q)$ is continuous, being a finite maximum of continuous functions. Its image is compact. Convexifying it in $\mathbb{R}^{|\Theta|}$ and applying the preceding finite-dimensional separation and affine-dependence argument proves (2.10) and its selector bound. Notice that the inner minimum is over stochastic q , not over deterministic tables: the nonlinear function e_θ does not commute with (2.13). If the attained visible optimum is zero, then for every u of positive weight and every θ one has $e_\theta(q^u) = 0$, since every summand is nonnegative. Any such q^u is a private exact simulator. The converse uses a selector of one value. This also proves the assertion for the optimized visible selector, whose support is finite by the bound just established.

Finally, cascade a K -state E -to- F transducer and an L -state F -to- G transducer. Within a step, consume the intermediate report before discarding it. The retained pair has at most $K_t L_t$ values. For any terminal tester on G , the second transducer together with that tester is an admissible randomized feedback system on F . The first error bound applies to this system, and total variation contracts under the remaining channel. The second error bound applies to the genuine F system. The triangle inequality gives (2.12). With selectors, condition on the independent selectors in the appropriate places; in the visible case keep both selectors in the compared transcript. The same argument is valid because the target reference selectors are independent. Storing a hidden selector

or exposing both visible selectors gives the stated costs. The minima established above may be used; equivalently one can use nearly optimal simulators and let the extra tolerance decrease.

A consumer is another causal transducer with task-specific readout and, where allowed, task-specific control, but with its required common encoder. The simulator is task blind, so its product with that encoder is still common. Condition on the consumer's independent random table to apply the feedback bound. Integration of any $[0, 1]$ marked loss changes by at most the total variation error. Taking the same paired consumer for every row proves the risk-set assertion, rather than a separate choice of simulator for each row. Infima of nonnegative weighted risks give the lower support inequalities. $\square$

Remark 2.3 (What the converse says). The dual (2.7) uses mixtures of indexed causal tests and optimizes over the entire common transducer. It is not a task-by-task choice of encoder. Its conclusion is an explicit hidden-selector simulation, with the selector subsequently charged if private storage is required. It does not replace the nonconvex private set by its convex hull at unchanged resource. The next example shows why that replacement would change the zero set.

2.3. A delayed channel and a strict signature separation. Let the source reveal $i \in [n]$ at checkpoint zero and reveal only a constant at checkpoint one. There is no control and the mark is constant. The target reports a constant at checkpoint zero and at checkpoint one draws $j \in [m]$ from a specified row-stochastic matrix $V = (V_{ij})$. The unknown parameter is i . The first target alphabet has one element: the simulator cannot use an early public output as unpriced storage. Only the intermediate register of size K matters.

Proposition 2.4 (The nonnegative-rank slice). *For this experiment,*

$$\delta_K^p(E, F_V) = \min_{A,B} \max_i \frac{1}{2} \sum_j |V_{ij} - (AB)_{ij}|, \quad (2.14)$$

where A is an n -by- K row-stochastic matrix and B is a K -by- m row-stochastic matrix. In particular

$$\delta_K^p(E, F_V) = 0 \iff \text{rank}_+(V) \leq K, \quad \delta_1^p(E, F_V) = 0 \iff V \text{ has identical rows.}$$

For $1 \leq K \leq n$ the identity channel has the sharp laws

$$\delta_K^p(E, F_{I_n}) = 1 - \frac{1}{\lceil n/K \rceil}, \quad \delta_K^{\text{mix}}(E, F_{I_n}) = 1 - \frac{K}{n}. \quad (2.15)$$

Thus, at tolerance $0 \leq \varepsilon < 1$, the least private width is $\lceil n/[(1-\varepsilon)^{-1}] \rceil$, whereas the least within-mode width with optimized hidden selection is $\max\{1, \lceil n(1-\varepsilon) \rceil\}$. The latter still pays for its selector. In particular, for $n = m = 3$ and $K = 2$,

$$\delta_2^p(E, F_{I_3}) = \frac{1}{2}, \quad \delta_2^{\text{mix}}(E, F_{I_3}) = \frac{1}{3}. \quad (2.16)$$

For $V_* = (1/2)I_3 + (1/6)\mathbf{1}\mathbf{1}^\top$,

$$\delta_{2,3}^h(E, F_{V_*}) = 0, \quad \delta_2^p(E, F_{V_*}) \geq \frac{1}{4\sqrt{3}} > 0, \quad \inf_r \delta_{2,r}^v(E, F_{V_*}) > 0. \quad (2.17)$$

Proof. The encoder maps the consumed i into its retained label by A ; the terminal decoder maps that label into j by B . This proves (2.14) including its converse implementation. If $V = CD$ is any nonnegative factorization, let $s_k = \sum_j D_{kj}$. Delete zero rows of D , define $B_{kj} = D_{kj}/s_k$ and $A_{ik} = C_{ik}s_k$. Since V is stochastic, the rows of A sum to one. Padding with unused states gives the claimed equivalence with nonnegative rank. The one-state statement follows immediately.

For $V = I_n$, row error is $1 - (AB)_{ii}$. If the least correct probability is c , for each i select a row $k(i)$ of B with $B_{k(i),i} \geq c$. Some row is selected at least $\lceil n/K \rceil$ times, so stochasticity gives $c \leq 1/\lceil n/K \rceil$. A partition into K groups of sizes at most $\lceil n/K \rceil$, each decoded uniformly, attains that bound. Also $\sum_i (AB)_{ii} \leq K$, by the calculation below with K in place of two. It remains

valid under mixing, giving hidden error at least $1 - K/n$. Start with a deterministic channel that reproduces K input symbols exactly and sends all others to those K outputs. Average all simultaneous permutations of its input and output labels. Every diagonal entry is then K/n , proving the second formula. Inverting the two monotone formulas gives the stated least widths.

For $V = I_3$, total variation in row i is $1 - (AB)_{ii}$. If all three diagonal entries of AB were greater than $1/2$, then for each i some row of B would have its i th entry greater than $1/2$. A stochastic row has at most one such entry, so two rows cannot serve three indices. This proves the private lower bound $1/2$. Merge two input symbols into a retained label that decodes uniformly to that pair, and retain the third symbol exactly. This attains it. For every stochastic factorization with two intermediate states,

$$\sum_{i=1}^3 (AB)_{ii} = \sum_{k=1}^2 \sum_{i=1}^3 A_{ik} B_{ki} \leq \sum_{k=1}^2 \sum_{i=1}^3 B_{ki} = 2.$$

The same inequality holds under hidden mixtures, giving maximal row error at least $1/3$. Mix uniformly the three preceding pair-merge channels. The resulting diagonal entries are $2/3$ and the off-diagonal entries are $1/6$. This is exactly V_* and has error $1/3$ from I_3 . It proves the hidden claims in both displays.

The smallest singular value of V_* is $1/2$. Any AB with two intermediate states has a unit vector x in its right null space, so $\|V_* - AB\|_F \geq \|(V_* - AB)x\|_2 \geq 1/2$. If the maximal row total variation error is ε , each row has Euclidean error at most 2ε , and hence the Frobenius error is at most $2\sqrt{3}\varepsilon$. This proves the explicit private bound. The optimized visible value is attained by Theorem 2.2; if it were zero, its zero-set assertion would contradict that bound. This proves its strict positivity. $\square$

2.4. Optimal resource sets and decision certificates. For a class $\mathcal{C}_{S,q}(F)$ of common consumers, with all other information conventions fixed, let $\mathcal{R}_{S,q}(F)$ be its vector of attainable $[0, 1]$ risks. For a nonnegative row weight w summing to one, write

$$h_F(S, q; w) = \inf_{r \in \mathcal{R}_{S,q}(F)} \langle w, r \rangle.$$

If E privately simulates F with cost K and error ε , then

$$\mathcal{R}_{S,q}(F) \subseteq \mathcal{R}_{KS,q}(E) + [-\varepsilon, \varepsilon]^J, \quad h_E(KS, q; w) \leq h_F(S, q; w) + \varepsilon. \quad (2.18)$$

For nonfinite row sets the first expression means pointwise paired risk functions within uniform distance ε ; no finite coordinate box is needed. Upper sets give the equivalent one-sided order formulation. Hidden selectors multiply private width by r ; visible selectors multiply the separate public alphabet by r . These are direct consequences of the final part of Theorem 2.2, with the optimal intrinsic error in place of an externally estimated simulator error.

When two stateless comparison directions have error at most ε , paired risk sets and their support values at unchanged width are within ε . If each is centered by a paired unrestricted benchmark whose error is also at most ε , the centered values are within 2ε . Therefore a gap $g > 0$ between zero excess and infeasible widths is stable under $\varepsilon < g/4$: a value below $g/2$ in the approximation certifies feasibility in the exact finite problem, and a value above $g/2$ certifies infeasibility. The benchmark and the class being optimized must be specified; this assertion does not identify different information signatures.

Proposition 2.5 (Intrinsic finite-dimensional selector approximation). *Let $\mathcal{R}$ be a compact risk image in $C(J)$, where J is compact. Suppose a bounded linear operator $L : C(J) \rightarrow C(J)$ has rank at most m and $\sup_{r \in \mathcal{R}} \|r - Lr\|_\infty \leq \eta$. Every barycenter $\bar{r}$ of a probability on $\mathcal{R}$ has a barycenter r_* of at most $m + 1$ members of $\mathcal{R}$ with $\|\bar{r} - r_*\|_\infty \leq 2\eta$. If $\mathcal{R}$ lies in a finite affine space of dimension*

m , the error is zero with at most $m + 1$ members. Thus the selector price can be controlled by the dimension of attainable risks, rather than by a covering number of the row space.

Proof. The compact set $L\mathcal{R}$ lies in an m -dimensional space. Its convex hull is compact, and $L\bar{r}$ belongs to it. The finite-dimensional affine-dependence argument in the proof of Theorem 2.2 gives $r_1, \dots, r_{m+1} \in \mathcal{R}$ and weights λ such that $L\bar{r} = \sum_i \lambda_i Lr_i$. Put $r_* = \sum_i \lambda_i r_i$. The two residual terms $\bar{r} - L\bar{r}$ and $r_* - Lr_*$ have norms at most η , proving the estimate. Apply the same argument after translation in the exact affine case. The implementation stores the selected design index whenever it is not provided as visible side information. $\square$

Remark 2.6 (Finite computation and general carriers). For rational finite instruments, (2.6) and the fixed- r formulations are finite polynomial feasibility problems; (2.7) is a finite linear program. These are exact variational characterizations, not assertions of polynomial running time in a succinct history description. The number of feedback trees, events, or deterministic tables can be large. On standard Borel carriers the same definitions use suprema over measurable feedback and tests and infima over executable kernels. Their product-state composition and decision transport remain valid by the same channel argument. Attainment and the finite testing dual are asserted here only for the stated finite class. The average-risk and physical results below give separate nonfinite theorems with their own hypotheses.

2.5. Local testing at the prescribed private budget. The convex converse in Theorem 2.2 has an explicit selector price. We now give a different converse that leaves the private register unchanged. Its witnesses are finite covers of the policy space by local testing inequalities, rather than one separating functional on a convex hull. Throughout this subsection the experiments, horizon, and alphabets are finite; for effective statements their instrument probabilities are rational. The prescribed widths and, when present, selector cardinality are fixed.

Let $\mathcal{P}$ be the product of the row simplexes of the private transducer. For a hidden or visible selector with at most r values, include its simplex and r copies of the transducer rows in $\mathcal{P}$. Zero selector weights are permitted. The relevant deficiency has the form

$$\delta = \min_{q \in \mathcal{P}} F(q), \quad F(q) = \max_{i \in I} f_i(q), \quad 0 \leq F \leq 1. \quad (2.19)$$

Here I consists of all parameter, feedback-tree, and marked-event tests. The empty event is included. For a visible selector a test specifies a feedback tree and event in each selector fiber. Thus

$$f_i(\lambda, q) = \sum_u \lambda_u \{Q_{\theta, g_u}(B_u) - P_{\theta, g_u}^{q_u}(B_u)\}.$$

For a hidden selector the target term is independent of λ . Every f_i is affine in each row separately, with rational coefficients. This follows by expanding complete controlled paths: a time-indexed row is used at most once along each path. It holds simultaneously for all feedback tests, not just for open-loop words.

For an integer $M \geq 1$, intersect a row simplex Δ_{d-1} with each closed coordinate box

$$\prod_{j=1}^d [b_j/M, (b_j + 1)/M], \quad b_j \in \{0, \dots, M - 1\}.$$

Discard empty intersections. Products of these row polytopes form a finite closed cover $\mathcal{C}_M$ of $\mathcal{P}$. Overlapping faces cause no ambiguity. Write $V(C)$ for the products of row vertices of a cell C . These are exactly its vertices. They are rational and are obtained by setting all but at most one coordinate of each row to a box endpoint and imposing that the coordinates sum to one. This

prescription also includes lower-dimensional cells and one-point rows. Define

$$\ell(C) = \max_{i \in I} \min_{v \in V(C)} f_i(v), \quad u(C) = \min_{v \in V(C)} F(v), \quad (2.20)$$

$$L_M = \min_{C \in \mathcal{C}_M} \ell(C), \quad U_M = \min_{C \in \mathcal{C}_M} u(C). \quad (2.21)$$

For time t , let d_t be the largest row alphabet cardinality, including both emitted symbol and next retained label. Put

$$\rho_t(M) = \begin{cases} 0, & d_t = 1, \\ \min\{1, d_t/(2M)\}, & d_t > 1, \end{cases} \quad b_M = 1 - \prod_{t=0}^T (1 - \rho_t(M)).$$

For a selector row set $s_M = 0$ if $r = 1$, and $s_M = \min\{1, r/(2M)\}$ otherwise. The three bounds used below are

$$\omega_M^p = b_M, \quad \omega_M^h = s_M + (1 - s_M)b_M, \quad \omega_M^v = \min\{1, 2s_M + (1 - s_M)b_M\}. \quad (2.22)$$

They do not count the number of histories or the number of policy rows. The visible-selector bound includes a second selector term because the reference joint experiment changes when its selector law changes.

Theorem 2.7 (Complete local testing certificates). *For any of the three fixed resource signatures in (2.19), the rational quantities (2.21) satisfy*

$$0 \leq L_M \leq \delta \leq U_M \leq 1, \quad U_M - L_M \leq \omega_M. \quad (2.23)$$

The upper witness is an executable rational transducer at the prescribed widths and selector cardinality. A lower witness consists of one test i_C for each cell and its inequalities $f_{i_C}(v) \geq L_M$ at all $v \in V(C)$.

For $M = 2^n$, the lower bounds increase and the upper bounds decrease to δ . In particular,

$$\delta = \sup_n \max_{(i_C) \in \mathcal{C}_{2^n}} \min_{C \in \mathcal{C}_{2^n}} \min_{v \in V(C)} f_{i_C}(v) = \inf_n U_{2^n}. \quad (2.24)$$

If $\delta > \varepsilon$, a finite local testing certificate eventually proves this strict inequality. If $\delta < \varepsilon$, a rational original-budget simulator eventually proves feasibility below ε . For rational ε , non-strict feasibility, including equality, is decidable by real quantifier elimination. The optimum is algebraic, and an optimal row table can be chosen with real algebraic entries.

Proof. Express each row of $q \in C$ as a convex combination of its row vertices. Expand a separately affine f_i in those combinations, one row at a time. The resulting weights on $V(C)$ are nonnegative and sum to one. Consequently

$$\min_{q \in C} f_i(q) = \min_{v \in V(C)} f_i(v). \quad (2.25)$$

It follows that $\ell(C) \leq \min_C F \leq u(C)$ and hence $L_M \leq \delta \leq U_M$. No assertion that F itself is separately affine has been made. This distinction is essential at a randomized optimum. The empty event gives $\ell(C) \geq 0$.

We next prove the oscillation estimate used for the gap. Two probability rows in the same M -cell differ in total variation by at most $\rho_t(M)$. Run the two transducers against the same tester and source. As long as all earlier emitted symbols and retained labels coincide, the controls and source laws agree. Maximally couple the next joint output–state row. The conditional chance of a mismatch is at most $\rho_t(M)$. Thus the chance of any mismatch through time T is at most b_M . On the event of no mismatch the source can be coupled identically, including its actual mark. This bounds the difference of the probability of every marked feedback event by b_M , uniformly in the parameter and tester. A finite causal instrument provides precisely the conditional kernels needed for this coupling.

For a hidden selector first maximally couple its two laws. The selector mismatch has probability at most s_M ; on matched fibers the preceding bound is b_M . The difference of any fixed test is at most $s_M + (1 - s_M)b_M$. There is only one executed fiber, so this estimate has no extra factor r . For a visible selector write its fixed test as the expectation of $a_u(q) = Q_{\theta, g_u}(B_u) - P_{\theta, g_u}^{q_u}(B_u)$. These numbers lie in $[-1, 1]$. On mismatched selector fibers their difference is at most two; on matched fibers it is at most b_M . Thus its oscillation is at most $2s_M + (1 - s_M)b_M$. This estimate does not forget the selector in either compared experiment.

Choose a vertex v_* minimizing F on $V(C)$ and an active test i_* there. For every other vertex,

$$f_{i_*}(v) \geq F(v_*) - \omega'_M = u(C) - \omega'_M,$$

where ω'_M is the untruncated corresponding oscillation bound. It follows that $u(C) - \ell(C) \leq \omega'_M$. Since both bounds lie in $[0, 1]$, truncation by one is valid. Choose a cell attaining L_M to obtain $U_M - L_M \leq \omega_M$. A vertex attaining U_M is an original row table; it is not a mixture of different designs.

Dyadic refinement only replaces each cell by subsets. By (2.25), its descendants cannot lower the minimum of a fixed test and hence cannot lower L_M . Every coarse vertex belongs to a finer cell and remains an extreme point of that subset; therefore the set of available vertices can only increase, and U_M cannot increase. The bounds in (2.22) tend to zero. Independent choices of the maximizing test in each of the finitely many cells give (2.24) and the two strict-decision assertions.

Finally $q \in \mathcal{P}$ and $f_i(q) \leq \varepsilon$ are finitely many polynomial conditions over the rationals. Quantifier elimination over real closed fields decides their existential conjunction; this is the classical effective Tarski–Seidenberg theorem, not a new elimination algorithm [11, §2.1, Theorem 2.1]. The set of feasible objective values is a semialgebraic ray with included endpoint δ . Its endpoint is algebraic over $\mathbb{Q}$. The nonempty optimum fiber has a point over the real algebraic numbers by the same real-closed-field transfer, proving the final assertion. $\square$

Corollary 2.8 (Private feasibility is locally test complete). *At fixed private widths, $\delta_K^P(E, F) \leq \varepsilon$ if and only if no finite dyadic local testing certificate has value greater than ε . For any finite prescribed list of resource profiles, all strictly feasible and strictly infeasible profiles can be separated by rational original-budget witnesses and finite local certificates. With rational tolerance, exact feasibility and the minimal elements in that finite list are decidable. These assertions require neither a free public selector nor a convex private behavior set.*

Proof. The first claim is (2.24). Apply the strict parts of Theorem 2.7 to each profile in the finite list and its quantifier-elimination clause to profiles on the boundary. The partial order is coordinatewise state and selector cost. Enumeration of feasible profiles then determines the minimal elements. It need not produce a single least profile, and no polynomial running time is asserted. $\square$

For the three-symbol channel V_* in Proposition 2.4, $\delta_2^{\text{mix}} = 0$ whereas $\delta_2^P \geq 1/(4\sqrt{3}) > 1/8$. Theorem 2.7 therefore eventually gives a finite local certificate exceeding $1/8$ at private width two. A single convex testing separator cannot have that conclusion, since the hidden convex hull already contains the target. This example identifies the logical work done by localization; it does not claim a polynomial bound on the size of the certificate.

The exact use of row vertices and real algebraic decision procedures is classical. So is the unrestricted filtered randomization criterion: The precise filtered antecedent is [10, Lemma 1 and Theorem 5], discussed in the introduction. The conclusion here is instead a complete system of local tests with executable witnesses at that fixed register, together with an explicit feedback modulus. These are different conclusions even on the finite delayed-channel slice. The original-source audit records separately the still-unverified comparison with Norberg’s full proof and with the original incomplete-machine minimization papers.

2.6. Transferring intrinsic certificates across presentations. For compatible marked experiments write $E \simeq_a \tilde{E}$ when two stateless, seed-free, marked causal channels, one in each direction, have feedback error at most a . Their report timing and action interfaces are the same. On a visible signature they compare the complete joint interface with the independent selector. These are assumptions about executable channels, not about unseeded marginal distributions.

Theorem 2.9 (Original-budget stability of intrinsic certificates). *Suppose $E \simeq_a \tilde{E}$ and $F \simeq_b \tilde{F}$. For private, fixed hidden-selector, and fixed visible-selector deficiencies at the same budget,*

$$|\delta_K(E, F) - \delta_K(\tilde{E}, \tilde{F})| \leq a + b. \quad (2.26)$$

If the approximating experiments are finite and rational, their local certificates consequently give the rigorous interval

$$\max\{0, L_M - a - b\} \leq \delta_K(E, F) \leq \min\{1, U_M + a + b\}. \quad (2.27)$$

An upper witness consists of the rational transducer and the two stateless bridge channels and has the original persistent resource budget. A lower certificate excludes every simulator at that budget for the original experiments whenever $L_M > a + b + \varepsilon$.

Proof. Compose $E \rightarrow \tilde{E}$, a near-optimal $\tilde{E} \rightarrow \tilde{F}$ transducer, and $\tilde{F} \rightarrow F$. The optimized composition inequality in Theorem 2.2, or its identical channel proof on measurable carriers, adds the three errors and multiplies the widths by $1, K, 1$. The bridges introduce no selector, so a prescribed hidden or visible cardinality remains unchanged. Reverse the bridges for the opposite inequality and take the infima. Equation (2.27) follows from Theorem 2.7. For the last assertion any original simulator would, after reverse bridging, contradict the finite lower certificate. Thus the lower statement does not assume the existence of an optimal simulator on a nonfinite carrier. $\square$

3. ENDOGENOUS COMMON DECISIONS

3.1. A shared encoder on a controlled tree. Let $\mathcal{H}_t$ be the finite supported observation–control histories of a known controlled experiment. The initial history has distribution μ , positive on $\mathcal{H}_0$. For $h \in \mathcal{H}_t$ and $a \in A_t$, write $p_t(y | h, a)$ for the next-report law and hay for the successor when this probability is positive. Zero-probability successors impose no condition. The finite task set is D . Task d has conditional expected stage costs $c_{t,d}(h, a)$ and terminal cost $c_{T,d}(h)$. These may be obtained by conditioning bounded losses of the actual hidden mark; the task does not alter the physical transition law. Controls do alter that law through a .

The common source-consuming encoder is

$$q_0(m | h), \quad q_{t+1}(m' | m, a, y), \quad m \in [S_t], \quad m' \in [S_{t+1}].$$

Controller d chooses its action by $\alpha_{t,d}(a | m)$. It cannot query the discarded history. The clock is available. The encoder does not know d ; the update sees the actual selected action, as allowed by the interface. All kernels can be stochastic. The same encoder must serve every task, even though different tasks induce different laws on histories.

Define the full-history Bellman functions and regrets by

$$\begin{aligned} V_{T,d}(h) &= c_{T,d}(h), \\ V_{t,d}(h) &= \min_{a \in A_t} \left(c_{t,d}(h, a) + \sum_y p_t(y | h, a) V_{t+1,d}(hay) \right), \\ G_{t,d}(h, a) &= c_{t,d}(h, a) + \sum_y p_t(y | h, a) V_{t+1,d}(hay) - V_{t,d}(h) \geq 0. \end{aligned} \quad (3.1)$$

The benchmark for task d is $v_d = \sum_h \mu(h) V_{0,d}(h)$. It is an ordinary full-history control value, not a posterior quantity conditioned on a fixed open-loop preparation.

Lemma 3.1 (Occupation regret). *For every admissible common encoder and task controller, the task excess satisfies the exact identity*

$$R_d - v_d = \mathbb{E}_d \sum_{t=0}^{T-1} G_{t,d}(H_t, A_t). \quad (3.2)$$

Moreover the entire vector of risks of a randomized finite-horizon design is a convex combination of risk vectors of deterministic designs with the same widths and one common encoder.

Proof. Take conditional expectations in (3.1) along the actual controlled path. Summing the resulting expressions for $G_{t,d}$ telescopes the terms $V_{t,d}(H_t)$, leaving the sum of stage and terminal costs minus $\mathbb{E}V_{0,d}(H_0)$. This proves (3.2) without assuming the control is open loop.

Sample in advance a deterministic outcome for every time-indexed row of the common encoder and every time-indexed row of every task controller. Row choices are independent with their prescribed probabilities; the sampled encoder table is shared across tasks. Along each trajectory each time-indexed row is encountered at most once. The product of its row probabilities is therefore the probability of that path under the sampled-table mixture. This identity holds for all tasks simultaneously after summing paths and costs. An additional independent visible or hidden seed is first conditioned on and then integrated. The conclusion is a mixture identity; the mixing variable is not claimed to be executable privately without retaining its index. $\square$

Definition 3.2 (Endogenous compatible cover). A cover of profile S consists of an initial map $i : \mathcal{H}_0 \rightarrow [S_0]$, common maps $\delta_t : [S_t] \times A_t \times Y_{t+1} \rightarrow [S_{t+1}]$, task action maps $a_{t,d} : [S_t] \rightarrow A_t$, and cells $C_{t,m}^d \subseteq \mathcal{H}_t$. Empty cells and overlapping cells are allowed. The requirements are

$$\begin{aligned} h &\in C_{0,i(h)}^d & (h \in \mathcal{H}_0, d \in D), \\ G_{t,d}(h, a_{t,d}(m)) &= 0 & (h \in C_{t,m}^d), \\ h \in C_{t,m}^d, p_t(y \mid h, a_{t,d}(m)) > 0 &\implies ha_{t,d}(m)y \in C_{t+1,\delta_t(m,a_{t,d}(m),y)}^d. \end{aligned} \quad (3.3)$$

Thus the cells cover the subtree generated by the selected actions. They need not cover histories generated by actions the controller never selects. Crucially, i and δ_t cannot depend on d .

Theorem 3.3 (Exact endogenous compression). *For a prescribed finite width profile S , the following are equivalent: a deterministic common design attains $R_d = v_d$ for every task; a privately randomized common design does so; a common design with an independent visible or hidden seed does so; and an endogenous compatible cover of profile S exists.*

Let $\beta_d > 0$ sum to one, and put $\mathcal{G} = \sum_d \beta_d (R_d - v_d)$. If the profile is infeasible, then every such design satisfies

$$\mathcal{G} \geq g_S := \beta_* \mu_* p_*^T \gamma_* > 0, \quad (3.4)$$

where $\beta_* = \min_d \beta_d$, μ_* is the least positive initial mass, p_* is the least positive transition probability, and γ_* is the least positive Bellman regret among supported entries. If there are no positive regrets, every profile with positive widths is feasible. The convention $p_* = 1$ applies when there are no transitions.

More generally, replacing the zero in (3.3) by the requirement $G_{t,d}(h, a_{t,d}(m)) \leq \eta_{t,d}$ gives a common design with $R_d - v_d \leq \sum_t \eta_{t,d}$. The sharper value is exactly the occupation-weighted sum in (3.2).

Proof. A cover defines the deterministic encoder $M_0 = i(H_0)$ and $M_{t+1} = \delta_t(M_t, A_t, Y_{t+1})$, and the task controller uses $a_{t,d}(M_t)$. Induction in (3.3) shows that every reached history lies in its indicated cell. Each incurred Bellman regret is zero, so Lemma 3.1 proves exactness. The same induction and identity prove the approximate-cover statement.

Conversely, an exact deterministic design defines $C_{t,m}^d$ to be the set of histories reached with positive probability under task d and state m . Initial coverage and common successor closure follow from its encoder. Since the nonnegative sum in (3.2) has zero expectation, every positively reached history has zero regret for its selected action. This proves all cover requirements. It also explains why adding closure for every unused action would impose a strictly different problem.

For a randomized design, Lemma 3.1 expresses the vector of excesses as an average of deterministic excess vectors. Each coordinate is nonnegative. If all averaged coordinates are zero, then, for almost every sampled design, all coordinates are zero, since the task set is finite. Choose one such deterministic table. This applies after conditioning on an arbitrary independent seed, with recovery from the seed and retained state when the seed is visible. It does not use a seed correlated with the initial history. The deterministic cover already constructed then proves necessity.

If no exact deterministic design exists, each deterministic design has some task, stage, and positively reached history with positive selected regret. That history has probability at least $\mu_* p_*^t$, and its regret is at least γ_* . Since $0 < p_* \leq 1$ and $t \leq T$, its contribution to $\mathcal{G}$ is at least the right side of (3.4). Every randomized design is an average of these vectors, so the same bound follows. When there is no positive regret, every action is Bellman optimal wherever supported, and a one-state encoder with arbitrary task actions is exact. $\square$

Example 3.4 (Why unused controls cannot define the cover). At the first time there is no report. Action **rest** leads to a deterministic hidden bit zero and a constant report, at zero cost. Action **probe** costs one, draws a fair bit, and reports it. After this report is consumed there is one blank checkpoint, followed by a binary guess with zero–one loss. The full-history optimum chooses **rest** and guesses zero; one retained state is exact. A restart family that insists on optimal guessing after the unused **probe** branch needs two retained states. The endogenous criterion and the all-controls restart criterion are both meaningful, but they have different feasible sets. The cover in Definition 3.2 implements the former, not the latter.

3.2. Lossy common design and intrinsic comparison. Let $\mathcal{D}_S^{\text{ctl}}$ be the finite set of deterministic common encoders and deterministic task controllers in this section. Put $r_d(\pi) = R_d(\pi) - v_d$. Then the hidden-selector lossy minimax value is

$$\min_{\lambda \in \Delta(\mathcal{D}_S^{\text{ctl}})} \max_d \sum_{\pi} \lambda_{\pi} r_d(\pi) = \max_{\beta \in \Delta(D)} \min_{\pi \in \mathcal{D}_S^{\text{ctl}}} \sum_d \beta_d r_d(\pi). \quad (3.5)$$

Here a selector chooses the entire design, including its task-controller tables; its mode is internal to that design. This follows from the simultaneous mixture identity in Lemma 3.1 and finite minimax. A mixture using at most $|D| + 1$ designs suffices and can be stored with at most $(|D| + 1)S_t$ private labels. Private finite-width optimization is instead the polynomial optimization in the actual encoder and controller rows. The exact zero sets agree by Theorem 3.3, but positive minimax values need not agree. Thus the cover theorem supplies a distinguished zero face inside a quantitative lossy theory, rather than purporting to solve every lossy design problem by equivalence classes.

Corollary 3.5 (Intrinsic certification of endogenous widths). *Suppose the losses of this finite problem are normalized so that every total task loss lies in $[0, 1]$. Let an approximation F have the same control and task interfaces, with each task represented by the same bounded marked-transcript loss on both sides and the full-history benchmark included in the paired decision class. If*

$$\max\{\delta_1^p(E, F), \delta_1^p(F, E)\} \leq \varepsilon < g_S/4,$$

then the optimized β -weighted excess at profile S in F is less than $g_S/2$ exactly when the endogenous cover exists in E . If only a cost- K simulation from E to F is known and an exact S -state F design transports with zero error and the same benchmark, then an endogenous cover of profile KS exists in E . A hidden selector multiplies that profile by its charged selector size.

Proof. Full-history controllers also transport through a stateless simulator, so their optimized benchmarks differ by at most ε . Equation (2.18) bounds the optimized constrained values by ε in each direction. The optimized excesses therefore differ by at most 2ε . In E the value is zero on the cover face and at least g_S otherwise, by Theorem 3.3. The strict threshold follows. For the one-sided statement, the product construction transports the exact design at the stated width and benchmark. The same theorem then converts exactness into a cover. $\square$

Remark 3.6 (Dependence of the finite witness). The number g_S depends on the least positive probability and regret of the specified finite model. It can tend to zero as a transition or a cost is perturbed. Neither (3.4) nor the certification corollary asserts a model-uniform robustness radius. Likewise an inverse-limit assertion that every finite prefix has a feasible cover would give existence of an infinite consistent design, not a computable bound on the first infeasible prefix. No such bound is used here.

4. REGENERATIVE AVERAGE RISKS

4.1. The reset signature. Let the physical state space X be standard Borel, with any actual persistent hidden mark included in X . Reports and actions are finite. A stationary common finite-register design consists of stochastic encoder rows and task-controller rows, independent of time; their product of simplexes is a compact set Π_S . A fixed finite task set is part of the row index $j \in J$, where J is compact metrizable. For each design and row, the joint physical-register transition has form

$$P_j^\pi = \eta N_j + (1 - \eta) Q_j^\pi, \quad 0 < \eta \leq 1, \quad N_j(x, m; \cdot) = \nu_j \otimes \delta_{m_0}. \quad (4.1)$$

The independent reset flag is physically supplied; it refreshes the physical preparation and sets the within-policy memory to m_0 . Thus N_j is a constant kernel independent of π . Actions between resets may change the physical state. No domination of Q_j^π by one reference measure is assumed.

Let $c_j^\pi : X \times [S] \rightarrow [0, 1]$ be the expected one-stage marked cost. We assume the following uniform continuity conditions, using the supremum over starting states for kernel total variation:

$$(j, \pi) \mapsto Q_j^\pi \text{ is continuous in } \sup_{x, m} \|\cdot\|_{TV}, \quad j \mapsto \nu_j \text{ in TV}, \quad (j, \pi) \mapsto c_j^\pi \text{ in } \|\cdot\|_\infty. \quad (4.2)$$

For fixed physical kernels with finite action and report alphabets, continuity in π follows directly by expanding the finite sums of encoder and controller probabilities; the physical kernels can be singular. Row dependence must independently satisfy (4.2). In particular changing a deterministic flow with the row does not automatically satisfy a total-variation continuity hypothesis.

Theorem 4.1 (Nonsummable average-risk duality). *Under (4.1)–(4.2), every (j, π) has a unique invariant probability ζ_j^π . Its average risk*

$$r_\pi(j) = \zeta_j^\pi c_j^\pi$$

is the limit of expected Cesàro average cost from every initial law. For $N \geq 1$ and $0 < \beta < 1$ the following errors are uniform in j, π , and the initial law ξ :

$$\left| \frac{1}{N} \sum_{t=0}^{N-1} \xi(P_j^\pi)^t c_j^\pi - r_\pi(j) \right| \leq \frac{1}{N\eta}, \quad (4.3)$$

$$\left| (1 - \beta) \sum_{t=0}^{\infty} \beta^t \xi(P_j^\pi)^t c_j^\pi - r_\pi(j) \right| \leq \frac{1 - \beta}{1 - \beta(1 - \eta)}. \quad (4.4)$$

The map $\pi \mapsto r_\pi$ is continuous into $C(J)$ and has compact image. The attainable average-risk body under independent selection of a stationary design is compact and convex in $C(J)$. For continuous

$b : J \rightarrow \mathbb{R}$ its common-design minimax value satisfies

$$\min_{\Lambda \in \mathcal{P}(\Pi_S)} \max_{j \in J} \left(\int r_\pi(j) d\Lambda(\pi) - b(j) \right) = \max_{w \in \mathcal{P}(J)} \min_{\pi \in \Pi_S} \int_J (r_\pi(j) - b(j)) dw(j), \quad (4.5)$$

and all displayed extrema are attained. For finite J a selector of at most $|J| + 1$ values realizes every required optimum. More generally Proposition 2.5 gives an intrinsic finite-dimensional selector approximation.

Proof. A probability kernel contracts total variation. Subtracting two probabilities in (4.1) cancels the common reset part, giving

$$\|\xi P_j^\pi - \xi' P_j^\pi\|_{\text{TV}} \leq (1 - \eta) \|\xi - \xi'\|_{\text{TV}}.$$

The explicit probability

$$\zeta_j^\pi = \eta \sum_{k=0}^{\infty} (1 - \eta)^k (\nu_j \otimes \delta_{m_0}) (Q_j^\pi)^k \quad (4.6)$$

is invariant by direct substitution, with convergence in total variation. Contraction proves uniqueness and $\|\xi(P_j^\pi)^t - \zeta_j^\pi\|_{\text{TV}} \leq (1 - \eta)^t$. Integrating the bounded cost and summing the geometric series proves (4.3) and (4.4), including their uniformity.

For two parameter-policy pairs let P, P' denote their kernels and ζ, ζ' their invariant laws. Invariance, the preceding contraction, and a one-step difference give

$$\|\zeta - \zeta'\|_{\text{TV}} \leq (1 - \eta) \|\zeta - \zeta'\|_{\text{TV}} + \sup_{x, m} \|P(x, m; \cdot) - P'(x, m; \cdot)\|_{\text{TV}}.$$

Thus the invariant-law difference is at most the last supremum divided by η . Together with the cost difference, this proves joint continuity of $(j, \pi) \mapsto r_\pi(j)$. Compactness of $J \times \Pi_S$ upgrades it to continuity of $\pi \mapsto r_\pi$ in the uniform norm. Its image is compact.

Probabilities on compact Π_S form a weakly compact set. The barycenter map into $C(J)$ is norm continuous: approximate the compact risk image uniformly by finitely many continuous coordinates, or use uniform equicontinuity of $r_\pi(j)$ and a finite net in J . The image is therefore compact and convex. The payoff $\int (r_\pi - b) d\Lambda dw$ is continuous and affine in both probabilities on compact spaces. Convex minimax gives (4.5). The inner minimum over a probability on Π_S equals the minimum over π , and compactness attains it. A finite-dimensional barycenter is reduced by the affine-dependence argument already proved in Theorem 2.2. This proves the finite selector assertion. $\square$

Remark 4.2 (Which policies are optimized). The theorem optimizes stationary finite-register common policies with the reset signature (4.1). It does not assert that a stationary policy is optimal among arbitrary nonstationary policies without that signature. A selector choosing a stationary policy is sampled once; its mode persists through resets while that mode's within-policy register is reset. If the selector must be private, store the mode and the register, paying their product. Conditioned on the mode, (4.3)–(4.4) hold with the same constants, so they also hold after averaging. The joint chain including a persistent mode need not have a unique invariant law across modes, and no such claim is needed.

4.2. Cycle comparison before optimization. Consider two reset experiments with the same reset probability and reset flag. A comparison within a fresh cycle starts with the specified physical preparation and initial register. Suppose, for every feedback design and every independent visible side seed, the joint laws of the marked information needed to evaluate the age- k stage cost differ by at most $\varepsilon_k \leq 1$. Here age zero means the first stage following a reset; its law includes any reports used at that stage. The same control law and encoder rows are used on both sides. This is a uniform channel comparison, not a comparison obtained after separately optimizing policies.

Theorem 4.3 (Regenerative resource transport). *The resulting paired average risks satisfy, for every consumer width,*

$$\|r_E - r_F\|_\infty \leq \mathfrak{e} := \eta \sum_{k=0}^{\infty} (1 - \eta)^k \varepsilon_k. \quad (4.7)$$

The bound holds with a priced simulator by replacing the consumer width S by its product with the simulator width and by charging the selector as in Theorem 2.2. For a sequence of stateless two-sided comparisons with $\varepsilon_{n,k} \rightarrow 0$ for each fixed k , $\mathfrak{e}_n \rightarrow 0$, uniformly over all consumer widths. At every width, the corresponding average-risk bodies and optimized lower support values have the same error bound. The minimax values in (4.5) with the same centering b differ by at most $\mathfrak{e}_n$.

Proof. Formula (4.6) expresses a stationary average risk as the geometric mixture of costs at each cycle age. The expectation of a $[0, 1]$ marked stage cost changes by at most ε_k in that age's joint law, yielding (4.7). The proof is uniform in the consumer because its error hypothesis is uniform before integration. For a priced transducer, include its retained state in the joint register and reset it to its specified initial state at the cycle boundary. The same product construction and conditioning on independent mode choices applies.

The geometric weights sum to one. Given $\delta > 0$, first choose L so their tail beyond L is less than $\delta/2$, and then choose n so the finite weighted sum through L is less than $\delta/2$. This proves $\mathfrak{e}_n \rightarrow 0$ without a rate uniform in k . Pairing each design with the same design on the other side gives the risk-body comparison. Taking weighted infima or the supremum of centered row risks proves the optimized statements; taking mixtures preserves the same bound. $\square$

Example 4.4 (A nondominated physical kernel). On $X = [0, 1]$, let $Q^a(x, \cdot) = \delta_{(x+a) \bmod 1}$ for finitely many rotations a , and let ν be Lebesgue probability. The transition $\eta\nu + (1 - \eta)Q^a(x, \cdot)$, as x ranges over X , is not dominated by a common σ -finite measure when $\eta < 1$: such a measure would have to give positive mass to uncountably many singleton points. A σ -finite measure has at most countably many positive-mass atoms, by decomposing into finite-measure sets and positive lower bounds on atomic mass. Finite report partitions and any common finite encoder fit Theorem 4.1. This is a nonsummable controlled risk theorem with a genuinely nondominated physical transition family, not an application of full-path density compactness.

Remark 4.5 (Why the reset is explicit). A hidden physical process may mix while a retained controller bit never changes. Physical mixing alone then gives no uniform contraction on the joint physical–register space. Equation (4.1) requires an actual clearing of that register. Conversely, without a cycle tail or another uniform long-time condition, finite-prefix comparison errors need not control average cost: a sequence of processes that agrees with a zero-cost process for its first n stages and incurs unit cost forever afterward has vanishing error at each fixed prefix but average cost one. The positive theorem uses the explicit geometric cycle law to control this precise obstruction.

5. AN ACTIVE MICROSCOPIC REALIZATION

5.1. Interventions and the unbounded flow. Fix a nonempty finite system of $N \geq 2$ identical hard spheres of positive diameter on a flat torus, with positive configuration volume. Let Γ_N be the admissible phase space and let μ be the normalized measure with density proportional to $\mathbf{1}_{\Gamma_N}(q, v) \exp(-b \sum_i |v_i|^2)$, $b > 0$. We use the classical almost-everywhere hard-sphere flow Φ_t , which is an invertible measure-preserving flow off a null exceptional set; see [6] and the detailed microscopic construction retained in the companion. No claim about particle-number uniformity is needed. On $H = L^2(\mu)$,

$$U(t)f = f \circ \Phi_t$$

is a strongly continuous unitary group, with skew-adjoint generator L . Indeed invariance gives isometry and invertibility; for bounded continuous phase functions, almost-everywhere continuity of the orbit at zero and dominated convergence give strong continuity. Density and isometry extend it to H . The generator domain includes the collision identifications of the flow; it is not the unrestricted formal transport derivative on arbitrary boundary data.

Let A be finite and choose orthogonal matrices O_a on velocity space. The intervention

$$R_a(q_1, \dots, q_N, v_1, \dots, v_N) = (q_1, \dots, q_N, O_a v_1, \dots, O_a v_N)$$

is an invertible μ -preserving map. Put $C_a f = f \circ R_a$. For a fixed step length $\Delta > 0$, the controlled physical step is $F_a = \Phi_\Delta \circ R_a$, and its Koopman operator is

$$T_a = C_a U(\Delta). \quad (5.1)$$

The order matters. If both $O_a = I$ and $O_a = -I$ are allowed, the two steps give different subsequent positions on a positive-measure set of configurations with no collision in the next Δ units of time and nonzero sufficiently small velocities. Such a set exists inside any nonempty interior admissible configuration region by choosing small velocities and then a positive interval away from zero. The action therefore changes the dynamics, rather than merely choosing which observable to read. We require no assertion that $C_a D(L)$ is contained in $D(L)$.

For all finite control words, the flow and intervention compositions are well defined on a common full-measure subset: the preimage of a null exceptional flow set under each finite measure-preserving composition is null, and there are countably many finite words. Every preparation absolutely continuous with respect to μ is supported there, including under a feedback choice of the word.

5.2. A graph-core approximation and controlled words. Choose an increasing family of finite-dimensional real subspaces $V_n \subset D(L)$ whose union is a graph core, and let P_n be the orthogonal projections. Such a family can be constructed without postulating smooth collision-boundary coordinates. Start with a countable dense family (f_j) in H and a smooth compactly supported approximate identity χ_ε on $\mathbb{R}$. Define

$$f_{j,\varepsilon} = \int_{\mathbb{R}} \chi_\varepsilon(s) U(s) f_j ds.$$

A difference quotient followed by translation of χ_ε shows that $f_{j,\varepsilon} \in D(L^k)$ for every k and

$$L^k f_{j,\varepsilon} = (-1)^k \int \chi_\varepsilon^{(k)}(s) U(s) f_j ds, \quad \|L^k f_{j,\varepsilon}\|_2 \leq \|\chi_\varepsilon^{(k)}\|_1 \|f_j\|_2. \quad (5.2)$$

For $f \in D(L)$ its smoothed versions converge to f in graph norm: commuting L on its domain reduces this to strong continuity for f and Lf . At each fixed smoothing scale, approximation of f by the dense family controls both the function and its first derivative by (5.2). A countable sequence of scales and finite linear spans consequently gives a graph core. Enumerate those vectors and let V_n be their successive spans. This also shows that their collision compatibility comes from the actual flow domain, not from an imposed incompatible trace.

Set $L_n = P_n L P_n$ on V_n , $U_n(t) = \exp(tL_n)$ on V_n , and, as an operator on H extended by zero on $V_n^\perp$, define

$$T_{a,n} = P_n C_a P_n U_n(\Delta) P_n. \quad (5.3)$$

These operators are contractions. Their finite-dimensional matrices need not be Markov matrices: they approximate *observables*; the observation experiment constructed below is nevertheless a genuine probability experiment.

Lemma 5.1 (Graph-core and controlled-word convergence). *For every $f \in H$ and $T_0 < \infty$,*

$$\sup_{0 \leq t \leq T_0} \|U_n(t) P_n f - U(t) f\|_2 \longrightarrow 0. \quad (5.4)$$

For every fixed word length k ,

$$\max_{(a_1, \dots, a_k) \in A^k} \|T_{a_1, n} \cdots T_{a_k, n} f - T_{a_1} \cdots T_{a_k} f\|_2 \longrightarrow 0. \quad (5.5)$$

Neither statement needs domain invariance under the interventions.

Proof. The matrix L_n is skew-adjoint on V_n because L is skew-adjoint and $V_n \subset D(L)$. Thus its group is unitary and $\|(z - L_n)^{-1}\| \leq 1/\Re z$ when $\Re z > 0$. If $u \in V_m$ and $n \geq m$, then

$$(z - L_n)^{-1} P_n (z - L) u = u.$$

The set $(z - L) \bigcup_m V_m$ is dense in H , since the union is a graph core and $z - L$ maps $D(L)$ onto H . The uniform resolvent bounds and $P_n \rightarrow I$ strongly therefore give

$$(z - L_n)^{-1} P_n f \longrightarrow (z - L)^{-1} f \quad (f \in H).$$

This is the graph-core consistency and stable-resolvent setting of Trotter's theorem [9, Theorem 5.2]. We give the contraction argument to specify the hypotheses used here. For any contraction generator A and $u \in D(A)$, the Euler resolvent formula can be written as a Gamma integral:

$$(I - tA/m)^{-m} u = \int_0^\infty e^{sA} u d\gamma_{m,t}(s), \quad \mathbb{E}_{\gamma_{m,t}} s = t, \quad \text{Var}_{\gamma_{m,t}} s = t^2/m.$$

This follows by multiplying the Laplace resolvent integrals and convolving their exponential densities. Since $\|e^{sA} u - e^{tA} u\| \leq |s - t| \|Au\|$, its error from $e^{tA} u$ is at most $t \|Au\|/\sqrt{m}$. For $u \in V_l$ one has $\|L_n u\| \leq \|Lu\|$ for all $n \geq l$. At each fixed $t > 0$ and Euler order m , the resolvent convergence already proved gives convergence of its m th power. First letting $n \rightarrow \infty$ and then $m \rightarrow \infty$ proves orbit convergence on the core. The orbit Lipschitz bound with constant $\|Lu\|$, uniform in n , upgrades it to uniformity on $[0, T_0]$ by a finite time mesh. Density and contraction extend it to all $f \in H$. No derivative of $U(t)f$ for $f \notin D(L)$ was taken.

Now $P_n C_a P_n \rightarrow C_a$ strongly by boundedness of C_a and strong convergence of P_n . Equation (5.4) gives $T_{a,n} \rightarrow T_a$ strongly. For a fixed finite word, subtract the two products one factor at a time. Each difference acts on a fixed vector formed from the limiting factors, while every factor to its left is a contraction. Each term tends to zero. There are finitely many words of length k , which proves the maximum in (5.5). At no step was an expression such as $LC_a f$ required. $\square$

5.3. Actual marked acquisition under feedback. Fix real observables $f_k \in H$ and noises $\sigma_k > 0$. At step k , a controller chooses a_k from its previously available reports and retained state, the action is applied to the physical state, and the new report is

$$Y_k = (T_{a_1} \cdots T_{a_k} f_k)(x_0) + \sigma_k \xi_k, \quad \xi_k \text{ independent standard Gaussians.} \quad (5.6)$$

The initial law is $p_\theta d\mu$, with $\sup_\theta \|p_\theta\|_2 \leq D < \infty$. An unobserved initial physical mark W is drawn jointly with x_0 through a fixed channel $\gamma(dw | x_0)$, independently of the acquisition noises. Losses may use that actual mark and the observed control-report transcript. This specification does not allow an unproved replacement of a controlled terminal physical quantity by an independent variable of the same marginal law.

The approximation replaces the product in (5.6) by $T_{a_1, n} \cdots T_{a_k, n} f_k$. It has a finite-dimensional physical realization. In a real orthonormal basis $(\varphi_1, \dots, \varphi_{d_n})$ of V_n , write the matrix of $T_{a,n}$ as $M_{a,n}$ and the coefficient vector of $P_n f_k$ as $c_{k,n}$. The initial row vector is $v_0 = (\varphi_1(x_0), \dots, \varphi_{d_n}(x_0))$, and

$$v_k = v_{k-1} M_{a_k, n}, \quad Y_k^{(n)} = v_k c_{k,n} + \sigma_k \xi_k.$$

Initialize (v_0, W) with the actual push-forward joint law of (x_0, W) . This is a valid controlled probability model with a real vector state; no positivity of $M_{a,n}$ is needed. Its real dimension d_n is not counted as d_n finite retained labels. The consumer register remains a separate constrained

object. Measurable representatives of the finitely many functions may be chosen once; their null ambiguities have zero probability under the specified preparations.

Theorem 5.2 (Active marked deficiency bound). *For each fixed horizon T , define*

$$\epsilon_{n,T} = 1 \wedge \frac{D}{\sqrt{2\pi}} \sum_{k=1}^T \frac{1}{\sigma_k} \sum_{a \in A^k} \|T_{a_1} \cdots T_{a_k} f_k - T_{a_1,n} \cdots T_{a_k,n} f_k\|_2. \quad (5.7)$$

The exact and approximate acquisitions have two-sided stateless marked causal deficiency at most $\epsilon_{n,T}$, using the identity report channel. The bound includes every independent visible seed in the joint comparison, holds uniformly over every finite-register consumer and feedback design, and tends to zero for each fixed T . Consequently all paired $[0, 1]$ task risks and optimized lower support values have error at most $\epsilon_{n,T}$ at unchanged widths; paired centered values have error at most $2\epsilon_{n,T}$ when their unrestricted benchmarks are also paired.

Proof. Couple the two experiments with the same x_0, W and every independent visible seed. Couple their private controller randomness so that the same available history produces the same action. At each step before the first report mismatch, use a maximal coupling of the two Gaussian reports. Gaussian densities satisfy

$$\|N(u, \sigma^2) - N(v, \sigma^2)\|_{\text{TV}} = 2\Phi(|u - v|/(2\sigma)) - 1 \leq \frac{|u - v|}{\sigma\sqrt{2\pi}}.$$

The equality follows by separating the densities at their midpoint; the inequality follows by bounding the standard normal density by $1/\sqrt{2\pi}$. Maximal couplings can be chosen measurably from these densities by their common part and residual parts.

For a fixed x_0 , the probability of a first mismatch at time k is bounded by the sum, over the possible matched action words $a \in A^k$, of their matched-history probabilities times the displayed Gaussian bound for that word. Discarding these probabilities gives an upper bound by the sum over all words. Integrate under the *original* preparation $p_\theta d\mu$. Cauchy–Schwarz bounds the integral of each absolute mean difference by D times its L^2 norm. Summing first-mismatch probabilities yields (5.7). In particular no bound on a feedback-conditioned posterior density has been assumed. When no mismatch occurs the entire transcript and actual mark, including the visible seed, coincide. The coupling inequality gives their joint total variation bound.

The same argument is symmetric in the two report models; the stateless identity channel is executable in both directions, so it bounds the intrinsic infima. Lemma 5.1 makes every term in the fixed finite sum tend to zero. The comparison precedes any consumer optimization, hence its bound does not depend on the consumer’s number of labels. The remaining conclusions are the channel and support inequalities in Theorem 2.2 and (2.18), which remain valid on these standard Borel carriers. $\square$

5.4. Active regenerative average decisions. Take a fixed observable f , fixed noise $\sigma > 0$, and a fixed finite measurable report partition applied to each Gaussian acquisition. At each cycle boundary the physical system is re-prepared from $p_\theta d\mu$ with its actual mark channel γ , and the within-policy register is reset to m_0 . Between boundaries the orthogonal interventions and hard-sphere flow continue as above. The independent reset probability is $\eta > 0$. In the approximate system reset the real vector and mark by the same joint push-forward preparation. A stationary consumer uses the partitioned reports, actions, and its finite register. Its $[0, 1]$ one-stage losses depend on the actual cycle mark, control, and current finite report; reset stage costs are fixed and identically paired. A finite task index is available to the controller but not to the common encoder.

Assume $j \mapsto p_j$ is continuous in $L^1(\mu)$ on a compact row set, the L^2 bounds are uniform, and the bounded row losses vary uniformly continuously. The flow, intervention maps, observable, noise law, report partition, and mark channel are fixed across rows. Then the exact and every approximate

model satisfy (4.2): only the reset law and losses vary with the row, and the dependence on finite policy rows is a finite polynomial expansion in kernels of norm one. The underlying physical transitions are allowed to be deterministic between resets.

Corollary 5.3 (Active average-risk realization). *Both models have the compact average-risk bodies and attained duality of Theorem 4.1. At every finite consumer width their paired average risks, lower supports, and equally centered minimax values differ by at most*

$$\epsilon_n = \eta \sum_{k=0}^{\infty} (1 - \eta)^k \epsilon_{n,k+1} \longrightarrow 0. \quad (5.8)$$

The convergence is uniform in the consumer width and in any independent priced mixture of stationary designs. It does not require a controlled word approximation rate uniform in word length.

Proof. A deterministic report partition contracts total variation. The age- k stage is evaluated after at most $k + 1$ controlled acquisitions, so Theorem 5.2 supplies the cycle comparison with $\varepsilon_{n,k} = \epsilon_{n,k+1}$. Apply Theorem 4.3. Its geometric tail proof gives the limit although the finite-word bound in (5.7) can grow with k . The reset and row-continuity assertions just verified give Theorem 4.1. Conditioning on each persistent mode and charging its index proves the mixture statement. $\square$

Remark 5.4 (Fixed physical scope). The interventions above are active changes to microscopic dynamics, and the average theorem uses actual repeated physical preparation. Neither observation control alone nor an assumed posterior L^2 bound is substituted for them. The assertions are for fixed N , sphere diameter, A , Δ , and positive noise, and for the declared initial-mark losses. No estimate is asserted uniform in particle number, a growing action alphabet, vanishing noise, or a kinetic scaling. The historical nonlinear kinetic corrector and the Sinai left-strip spectral problem remain different analytic statements. They are preserved in the development with their original hypotheses, not inferred from the present operator approximation.

5.5. A posteriori bounds from graph and intervention residuals. Strong controlled-word convergence supplies an asymptotic comparison. For a fixed approximation space one can also bound its actual comparison error using finite Gram data. We retain the physical setup of Theorem 5.2: the particle number, interventions, noise, step length, and actual initial-mark channel are fixed. These statements do not require the interventions to map $D(L)$ into itself.

Suppress n and let $V \subset D(L)$ have orthonormal basis $\phi_1, \dots, \phi_d$. Identify V with $\mathbb{R}^d$ by an isometry ι ; complexification gives the same formulas with adjoints. Write

$$P = \iota \iota^*, \quad J = \iota^* L \iota, \quad B_a = \iota^* C_a \iota, \quad M_a = B_a e^{\Delta J}.$$

Because L is skew-adjoint and C_a is unitary, J is skew-adjoint, e^{sJ} is unitary and M_a is a contraction. The two residual Gram matrices are

$$R = (L \iota)^* L \iota - J^* J = ((I - P) L \iota)^* ((I - P) L \iota), \quad (5.9)$$

$$S_a = I - B_a^* B_a = ((I - P) C_a \iota)^* ((I - P) C_a \iota). \quad (5.10)$$

In particular both are positive semidefinite. For a coefficient vector c , set

$$d_a(c) = \int_0^\Delta \sqrt{(e^{sJ} c)^* R (e^{sJ} c)} ds + \sqrt{(e^{\Delta J} c)^* S_a (e^{\Delta J} c)}. \quad (5.11)$$

All quantities in this formula are finite-dimensional. For a real observable $f \in H$ put

$$c_f = \iota^* f, \quad \tau_f = (\|f\|_H^2 - \|c_f\|^2)^{1/2}.$$

If $w = (a_1, \dots, a_k)$ and $c_j = M_{a_{j+1}} \cdots M_{a_k} c_f$ (empty product equal to the identity), define

$$E_w(f) = \tau_f + \sum_{j=1}^k d_{a_j}(c_j). \quad (5.12)$$

The suffix order agrees with the Koopman product for a physically chronological word; its reversal would approximate a different dynamics.

Theorem 5.5 (Residual control of the active marked experiment). *For every such space V and every controlled word,*

$$\|T_{a_1} \cdots T_{a_k} f - \iota M_{a_1} \cdots M_{a_k} c_f\|_H \leq E_w(f). \quad (5.13)$$

For the preparations p_θ with $\sup_\theta \|p_\theta\|_2 \leq D$ and Gaussian acquisitions of Theorem 5.2, the two stateless marked feedback deficiencies are bounded by

$$\widehat{\epsilon}_{V,T} = \min \left\{ 1, \frac{D}{\sqrt{2\pi}} \sum_{k=1}^T \frac{1}{\sigma_k} \left(\sum_{w \in A^k} E_w(f_k)^2 \right)^{1/2} \right\}. \quad (5.14)$$

The comparison includes every independent exposed seed and is uniform over all finite consumer widths. Thus it bounds raw paired risks and lower supports at unchanged widths. When unrestricted benchmarks are also paired, the centered risk bound is $2\widehat{\epsilon}_{V,T}$.

Proof. The orthogonal splitting of $L\iota c$ into its V component ιJc and its orthogonal component proves (5.9). Unitarity of C_a similarly proves (5.10). For $v = \iota c$, the path $s \mapsto \iota e^{sJ} c$ lies in the finite space $V \subset D(L)$ and is differentiable in the graph norm. Hence variation of constants is legitimate and gives

$$U(\Delta)v - \iota e^{\Delta J} c = \int_0^\Delta U(\Delta - s)(L\iota - \iota J)e^{sJ} c ds.$$

Taking norms and applying (5.9) bounds this difference by the integral in (5.11). Apply the unitary C_a and then add the orthogonal projection error $\|(I - P)C_a \iota e^{\Delta J} c\|$ to obtain

$$\|T_a \iota c - \iota M_a c\| \leq d_a(c).$$

No derivative of $C_a \iota c$ has occurred. First remove $(I - P)f$ at cost τ_f . Telescope the difference of the two word products with exact prefixes and approximate suffixes. All prefixes are isometries and all approximate suffixes are contractions. The j th term is bounded by $d_{a_j}(c_j)$, proving (5.13).

For the probabilistic assertion use the common initial x_0 , its actual conditional mark W , and the same visible seed, as in Theorem 5.2. Before the first report mismatch the selected actions agree. Conditional on x_0 , the weights of the matched histories leading to the possible words of length k sum to at most one. If $e_w(x)$ is the corresponding exact-minus-approximate mean, the first-mismatch contribution at time k is therefore at most

$$\frac{1}{\sigma_k \sqrt{2\pi}} \int p_\theta(x) \max_{w \in A^k} |e_w(x)| d\mu(x).$$

This retains the history weights until after taking their maximum, instead of replacing each by one. Cauchy–Schwarz and $\max_w |e_w|^2 \leq \sum_w |e_w|^2$ bound the integral by $D(\sum_w \|e_w\|_2^2)^{1/2}$. Apply (5.13) and sum the first-mismatch probabilities. The coupling is symmetric and preserves the whole actual marked transcript on its matching event, so it proves both stateless inequalities in (5.14). It conditions only on the original preparation, not on a feedback posterior with a supposed L^2 bound. Conditioning on an independent exposed seed gives the same uniform estimate. Risk and support statements follow by composition with any consumer and by taking the appropriate infima. $\square$

Corollary 5.6 (Finite-data average-risk and presentation certificates). *In the active regenerative model of Corollary 5.3, a finite age cutoff H gives an error bound for paired average risks and lower supports, uniform in consumer width,*

$$\hat{e}_{V,H} = \eta \sum_{k=0}^H (1-\eta)^k \hat{e}_{V,k+1} + (1-\eta)^{H+1}. \quad (5.15)$$

The same bound applies to identically centered values; separately paired unrestricted benchmarks give twice the bound. Finite-horizon bounds (5.14), combined with any other verified stateless presentation bridges, can be inserted into (2.27) without changing the intrinsic resource budget.

Proof. At age k there are at most $k+1$ acquisitions. Use Theorem 5.5 at those horizons and bound each remaining age error by one. The remaining geometric mass is exactly $(1-\eta)^{H+1}$. The cycle formula in Theorem 4.3 proves (5.15). The final assertion is Theorem 2.9. A finite-dimensional vector model is not automatically a finite report experiment; a further finite-report bridge must actually be proved before invoking the rational local testing algorithm. No such bridge is implicit in a matrix representation. $\square$

The data required in (5.9)–(5.12) are inner products of $f, \phi_i, L\phi_i, C_a\phi_i$. Once these data have rigorous enclosures, the formulas give a posteriori bounds without knowing the exact controlled orbit. For example, certified numbers $r_0 \geq \|R\|$ and $s_{a,0} \geq \|S_a\|$ imply the simpler bound

$$d_a(c) \leq (\Delta\sqrt{r_0} + \sqrt{s_{a,0}})\|c\|.$$

With matrix-entry enclosures, interval arithmetic and this inequality avoid an uncertified numerical integral. Floating-point Gram entries alone are not an analytic certificate. Nor is it asserted that these residual majorants converge for every graph-core sequence: cancellation may make the actual orbit error small while its residual bound is large. The convergence theorem remains Lemma 5.1; Theorem 5.5 is a complementary finite-space bound. This distinction separates a proved limiting approximation from a checkable error estimate for a particular space.

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